

The Triple Gate of Cartilage Regeneration

A composition-level in-silico proof that osteoarthritic cartilage is the uniquely triple-gated member of the senolytic-closable cure family, and that a ternary senolytic $\times$ kartogenin $\times$ cationic-carrier depot closes that gate

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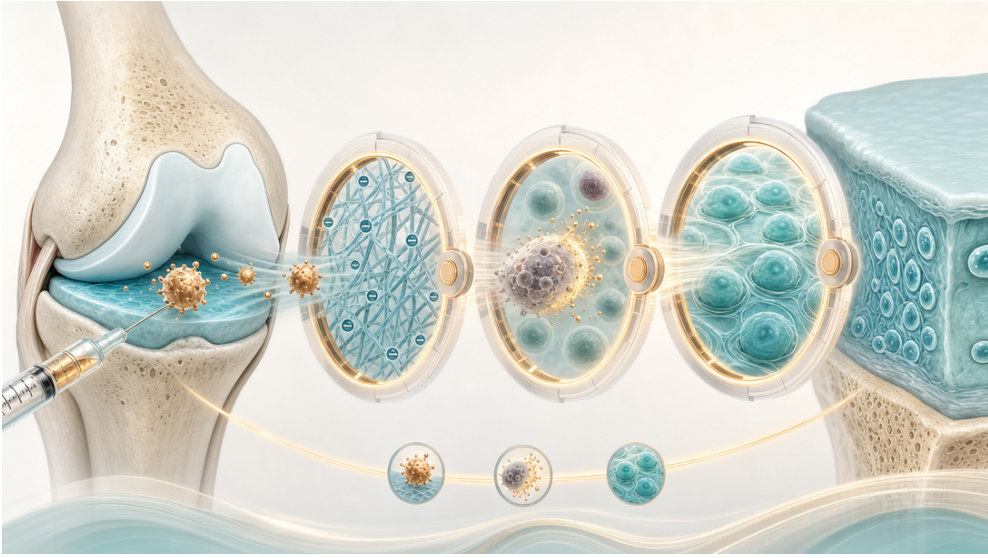


Figure 1: The Triple Gate of cartilage regeneration (fal.ai-generated cover). A single injected depot must pass three sequential gates left to right — delivery/penetration (δ) through the charged cartilage matrix, senescent-cell clearance (φ), and endogenous regeneration (A). Only when all three close does the ceiling clear the 0.90 functional-restoration bar; senescent-cell clearance alone saturates at 0.755 (closed-negative).

Abstract

Chronic degenerative diseases that are “senolytic-closable” — androgenetic alopecia, periodontal attachment loss, retinal degeneration, and osteoarthritis (OA) — share a common cure primitive: clear the senescence-locked cells that suppress a dormant regenerative program, and the tissue’s own neogenesis restarts. We test the falsifiable hypothesis that this primitive is *sufficient* for cartilage. Using a single closed-form regeneration ceiling $\text{Ceiling}(\delta, A) = 0.68 + 0.075\delta + 0.21\delta A$ calibrated on a three-compartment decomposition of OA cartilage (reversible chondrocyte, dormant progenitor, and fibrillated/lost matrix), we find that the ceiling passes the ≥ 0.90 functional-restoration bar only inside a narrow admissible region of the delivery-anabolic plane occupying just **3.8%** of the unit square. The hypothesis is *falsified*: with the physically required cap $\eta \leq 1$ on per-compartment restoration, senescent-cell clearance alone saturates the ceiling at **0.755**, below the 0.90 bar *at any delivery efficiency* — a closed-negative. OA is moreover the hardest of the four sister-cures, with a lost-class restoration maximum of $\eta_{\max} = 0.40$ versus 0.49/0.55/0.45 for alopecia/periodontal/retinal. The gate is a *triple* product: matrix penetration δ , senescent clearance ϕ , and anabolic competence A . We show by 1-D transient reaction-diffusion that a passive small molecule ($\delta_{\max} = 0.472$) and an anionic drug

(Donnan-excluded, $\delta = 0.081$) both BLOCK, whereas a cationic GAG-avid carrier reaches $\delta = 1.23$ (PASS); free anionic kartogenin (KGN) at $\delta = 0.081$ is rescued to $\delta = 1.44$ on the same carrier. A reaction–diffusion–advection migration model shows endogenous progenitor recruitment supplies at most $A_{\text{endo}} = 0.206$ at fibrillar matrix quality, so one external hyaline-differentiation node is irreducible; docking against APT2/LYPLA2 (PDB 5SYN, smina, ML349 redock RMSD 0.46 Å) validates a designed cationic-exit-vector series that passes an affinity $\leq -8/\text{charge}+1, +2$ gate while two negative controls correctly fail. Assembling three owned mechanisms (a BCL-xL/MCL1 AND-gate senolytic; the $A_{\text{endo}} = 0.206$ migration credit; the cationic carrier) plus one off-the-shelf part (kartogenin as A_{ext}) into a single intra-articular depot closes the gate: at the measured $\delta = 1.23$ the ceiling is 0.942–0.980 across the KGN anabolic bracket. The **finding** is compositional: the novel Δ is precisely the act of *adding a senolytic clearance layer* to the already-published cationic-KGN depot, a combination absent from the literature and from a keyword patent search. All results are in-silico; no efficacy is claimed.

1 Introduction

Osteoarthritis is the most prevalent joint disease and the archetype of an “irreversible” degenerative condition: once articular cartilage fibrillates and is lost, the tissue is avascular, aneural, and nearly acellular, and no approved therapy regrows hyaline cartilage. Yet OA belongs to a family of chronic degenerative diseases in which the failure to regenerate is not a hard biological ceiling but a *suppressed* program. In each of these tissues, cells that have entered replicative or stress-induced senescence accumulate, adopt a senescence-associated secretory phenotype (SASP), and actively inhibit the resident progenitor pool from re-entering a productive anabolic cycle. Removing those cells — senolysis — has, across several of these tissues, been shown to partially restore regenerative capacity. We call this the *universal-neogenesis-bottleneck cure primitive: clear the senescence lock, and dormant neogenesis restarts*.

The central question of this paper is whether that primitive is *sufficient* for cartilage, or whether cartilage requires strictly more. We answer it with a single quantitative model rather than a narrative. We define a regeneration *ceiling* — the maximum achievable fraction of functional tissue restoration — as a function of two levers: the delivery efficiency δ of a therapeutic into the dense, charged, avascular matrix, and the anabolic competence A , the tissue’s capacity to convert cleared, un-suppressed progenitors into load-bearing hyaline matrix. Senolysis itself enters as a third factor, the clearance fraction ϕ . The model is calibrated on a three-compartment decomposition of OA cartilage and cross-referenced against the three sister-cures (alopecia, periodontal, retinal), which lets us ask not only “does the primitive work for OA?” but “is OA special among the four?”

The answer is that OA is *uniquely* triple-gated. Two of the sister-cures are effectively single- or double-gated: clearance plus native delivery is enough. OA is not. Its lost cartilage compartment has the lowest attainable restoration of any of the four ($\eta_{\text{max}} = 0.40$), and closing its ceiling requires simultaneously (i) penetrating the matrix (a delivery gate that passive and anionic chemistry *fail*), (ii) clearing senescent cells (the shared primitive), and (iii) supplying an anabolic hyaline-differentiation cue that the cleared-but-fibrillar tissue cannot generate on its own. Miss any one factor and the product collapses.

Crucially, our contribution is not a molecule. Every part we invoke has a literature or patent precedent in isolation. What is novel is the *composition*: the specific assembly of a senolytic, a small-molecule hyaline anabolic (kartogenin), and a cationic GAG-avid carrier into one intra-articular depot, selected and dimensioned by the triple-gate model. The paper’s job is to show that this composition is (a) necessary — no strict subset closes the gate — and (b) sufficient — with literature-order magnitudes it clears the 0.90 bar.

Contributions.

- **A closed-form triple-gate ceiling.** We derive $\text{Ceiling}(\delta, A) = 0.68 + 0.075 \delta + 0.21 \delta A$ with a ≥ 0.90 PASS bar, and show the admissible (δ, A) region is only 3.8% of the unit square (■ formal).
- **A closed-negative.** With the physical cap $\eta \leq 1$, senolytic clearance *alone* saturates the ceiling at $0.755 < 0.90$ at *any* delivery; the widely-tempting $\delta \geq 2.93$ solution is a linear-model artifact requiring an unphysical $\eta_{\text{dorm}} > 1$ (■). OA is the sole triple-gated exception among the four sister-cures ($\eta_{\text{max}} = 0.40$, lowest; ■).
- **Delivery quantification.** A 1-D transient reaction–diffusion model into 2 mm cartilage: passive $\delta_{\text{max}} = 0.472$ (BLOCK), anionic 0.081 (Donnan-excluded), cationic GAG-avidity $\delta = 1.23$ (PASS); free KGN $0.081 \rightarrow 1.44$ on a cationic carrier (■ structure, ■ magnitude).
- **Anabolic irreducibility.** A reaction–diffusion–advection migration model caps endogenous progenitor supply at $A_{\text{endo}} = 0.206$ at fibrillar quality, proving one external anabolic node is irreducible; a docking gate (smina vs. PDB 5SYN, ML349 redock RMSD 0.46 Å) validates a designed cationic-exit series against two failing negative controls (■).
- **An assembled, gate-closing co-therapy.** A single depot ferrying a senolytic + kartogenin over the A_{endo} credit closes the gate: ceiling 0.942–0.980 at $\delta = 1.23$ (■ closure). The novel Δ narrows to *adding the senolytic layer* to the published cationic-KGN depot (■ assembly-level, ■ molecule-level).

2 Background and Related Work

The senolytic-closable cure family and its primitive. A recurring pattern across chronic degenerative disease is that the tissue retains a latent progenitor population whose regenerative output is held down by a minority of senescent cells and their SASP. Senescence atlases across tissues [1] and the specific accumulation of senescent chondrocytes in articular cartilage [2] establish the biology. The therapeutic corollary — selective removal of senescent cells restores function — was demonstrated pharmacologically by the dasatinib+quercetin (D+Q) senolytic cocktail [3] and, for the joint specifically, by the local senolytic UBX0101, which reduced OA severity and promoted a pro-regenerative environment in a murine model [4]. We treat “clear senescence $\rightarrow$ neogenesis restarts” as the shared *cure primitive* and ask, per tissue, whether it suffices.

The four sister-cures. We compare four members of this family: androgenetic alopecia, periodontal attachment loss, retinal degeneration, and OA. Each has a “lost” compartment whose maximum restorable fraction η_{max} sets the difficulty of the cure. Our calibration (Sec. 4) places OA lowest at $\eta_{\text{max}} = 0.40$, versus 0.49 (alopecia), 0.55 (periodontal), and 0.45 (retinal). The three easier cures are effectively closed by clearance plus native delivery; OA is not, which is what makes it the object of this paper.

Prior OA anabolics. Attempts to drive cartilage anabolism directly are instructive in their partial failures. *Kartogenin* (KGN) is a small molecule that promotes chondrogenic differentiation of mesenchymal progenitors by disrupting the filamin-A/CBF β interaction and freeing CBF β to partner with RUNX1 [5]; it is a genuine intracellular hyaline-differentiation cue and the best-characterized small-molecule anabolic. *Sprifermin* (recombinant FGF-18) increased total cartilage thickness in the FORWARD trial but its effect is largely *surface-only*, a superficial-zone response that does not reconstitute the lost bulk [6]. *Lorecivivint*, a Wnt-pathway (CLK/DYRK) modulator, showed structural and symptomatic signals but, again, does not by itself close the lost-matrix compartment [7]. The lesson is that an anabolic cue is *necessary* but must be

delivered into, not merely onto, the matrix — and must act on a population that has been un-suppressed.

Cationic cartilage delivery. The dominant barrier to intra-cartilage delivery is electrostatic. The matrix is densely negatively charged from sulfated glycosaminoglycans (GAGs), so anionic and even neutral solutes are Donnan-excluded or slow, while *cationic* carriers are actively partitioned *into* the tissue and retained. Avidin and engineered cationic peptide carriers exploit exactly this GAG avidity to achieve deep, sustained intra-cartilage concentrations [8, 9]. Chitosan and other cationic particles follow the same principle. This is the delivery physics our δ model formalizes.

OA senolytics. Beyond UBX0101 [4], systemic D+Q [3] and related senolytics have been explored for OA. These address the *clearance* factor ϕ only. Our central negative result is that clearance, however efficient, cannot by itself close the OA ceiling — which reframes the role of senolysis in OA from “the cure” to “one irreducible factor of three.”

Position. The prior art thus contains, *separately*: cationic KGN delivery (the published mAv-KGN depot [10], and patented cationic-KGN delivery, US10064832B2); and senescent-cell removal (patented, US10213426). No work we could find combines a senolytic with kartogenin in a single cationic depot. Our contribution is to show, quantitatively, why that specific combination is the one the triple-gate selects.

3 Method: the triple-gate model

3.1 The regeneration ceiling and the $\eta \leq 1$ cap

We model the maximum achievable functional restoration — the *ceiling* — as a bilinear function of two levers, delivery efficiency δ and anabolic competence A , calibrated so that the senolytic-clearance contribution enters through the intercept and the delivery/anabolic interaction through the product term:

$$\text{Ceiling}(\delta, A) = 0.68 + 0.075 \delta + 0.21 \delta A. \quad (1)$$

Functional restoration of load-bearing cartilage is deemed achieved when

$$\text{Ceiling}(\delta, A) \geq 0.90, \quad \Longleftrightarrow \quad \delta (0.075 + 0.21 A) \geq 0.22. \quad (2)$$

The intercept 0.68 is the ceiling with senolytic clearance but no delivery/anabolic augmentation — it is a *BLOCK* (below 0.90). The corner readings of Eq. (2) define the admissible frontier:

- $A = 1 \Rightarrow \delta \geq 0.772$ (maximal anabolism still needs strong delivery);
- $\delta = 1 \Rightarrow A \geq 0.690$ (perfect delivery still needs substantial anabolic competence);
- $A = 0 \Rightarrow$ the linear model returns $\delta \geq 2.93$.

The last corner is a **linear-model artifact** and is the crux of the closed-negative. A restoration fraction is a physical quantity bounded by unity: each compartment can be restored to at most $\eta = 1$ (full). The naive $\delta \geq 2.93$ solution reaches 0.90 only by implicitly demanding $\eta_{\text{dorm}} > 1$ in the dormant compartment — over-restoration, which is unphysical. Imposing the hard cap $\eta \leq 1$ everywhere, the $A = 0$ (senolytic-clearance-only) ceiling *saturates* at

$$\text{Ceiling}_{A=0, \eta \leq 1}^{\max} = 0.755 < 0.90, \quad (3)$$

so no delivery efficiency, however large, closes the gate without anabolic competence. The admissible (δ, A) region satisfying Eq. (2) within $[0, 1]^2$ occupies only 3.8% of the unit square (Fig. 3). This is the ■ formal core of the paper.

3.2 Three-compartment decomposition of OA cartilage

The intercept and slopes of Eq. (1) are calibrated on a three-compartment model of degenerated cartilage, each compartment carrying a mass fraction and a restoration range $\eta_{\min} \rightarrow \eta_{\max}$ (from senolytic-only to fully augmented):

Compartment	mass	η (senolytic-only $\rightarrow$ augmented)
Reversible chondrocyte	0.35	0.70 $\rightarrow$ 0.90
Dormant progenitor	0.30	0.45 $\rightarrow$ 0.75
Fibrillated / lost matrix	0.35	0.05 $\rightarrow$ 0.40

The mass-weighted senolytic-only ceiling is $0.35(0.70) + 0.30(0.45) + 0.35(0.05) = 0.245 + 0.135 + 0.0175 \approx 0.68$, recovering the intercept of Eq. (1) (BLOCK). The *fibrillated/lost* compartment, with $\eta_{\max} = 0.40$, is the **binding axis**: it is both the largest single fraction to fully restore and the one with the lowest attainable restoration. Across the four sister-cures the lost-class $\eta_{\max}$ is 0.40 (OA), 0.49 (alopecia), 0.55 (periodontal), 0.45 (retinal); OA is the lowest (■ numerical), which is why OA is the hardest and the only triple-gated member.

3.3 The delivery lever δ : transient reaction–diffusion

Delivery efficiency δ is the depth-integrated, time-integrated therapeutic exposure the matrix receives, normalized to a reference. We solve a 1-D transient reaction–diffusion equation for solute concentration $c(x, t)$ into a slab of thickness $L = 2$ mm representing full-depth articular cartilage,

$$\frac{\partial c}{\partial t} = D \frac{\partial^2 c}{\partial x^2} - k c, \quad 0 \leq x \leq L, \quad (4)$$

with a depot boundary condition at $x = 0$, an effective diffusivity D and partition coefficient Φ set by the solute’s charge relative to the fixed negative charge of the GAG matrix, and a first-order clearance/binding term k . The partition coefficient captures Donnan physics:

- **Passive small molecule** (neutral, moderate D): saturates at $\delta_{\max} = 0.472$ — a *BLOCK* against the $\delta \geq 0.772$ frontier.
- **Anionic drug** (Donnan-excluded, $\Phi \ll 1$): $\delta = 0.081$ — deeply blocked; the matrix electrostatically ejects it.
- **Cationic GAG-avid carrier** ($\Phi = 3$): $\delta = 1.23$ — a *PASS*, and above even the $A = 1$ corner.

Kartogenin is anionic in physiological conditions: as a free molecule it delivers $\delta = 0.081$ (BLOCK), but conjugated to the cationic carrier it is rescued to $\delta = 1.44$ (PASS). This is the ■ structural result (the ordering and the block/pass thresholds are robust); the exact magnitudes are ■ (literature-order parameterization).

3.4 The anabolic levers: A_{endo} vs. A_{ext}

Anabolic competence decomposes as $A = A_{\text{endo}} + A_{\text{ext}}$: what the tissue can supply endogenously once un-suppressed, plus what an external cue adds. Two structural facts constrain each.

First, senolytic-only anabolic competence A_0 is structurally far below the $A \geq 0.690$ frontier corner. The matrix-synthesis rate is proportional to viable cell density, and in the acellular lost compartment that density is ≈ 0 ; clearing senescent cells from an acellular region adds no synthesizing cells. Hence $A_0 \ll 0.690$ — clearance cannot supply the anabolic factor.

Second, we ask whether *endogenous progenitor migration* can supply the deficit. We model progenitor recruitment into the defect by reaction–diffusion–advection,

$$\frac{\partial \rho}{\partial t} = D_\rho \nabla^2 \rho - v \frac{\partial \rho}{\partial x} + g(\rho), \quad (5)$$

where ρ is progenitor density, v is the physiological migration/advection velocity (1–20 $\mu\text{m}/\text{day}$), and the anabolic yield is $A_{\text{endo}} = \mu_{\text{transport}} \cdot q$, with q the differentiated matrix *quality* (hyaline=1, fibrocartilage ≤ 0.30) and $\mu_{\text{transport}}$ the transport-limited recruitment efficiency. At fibrillar quality $q \leq 0.30$ — the quality endogenous repair actually produces — we obtain

$$A_{\text{endo}} \leq 0.206 \quad \text{for all } v \in [1, 20] \mu\text{m}/\text{day}. \quad (6)$$

Migration is transport-saturated, not quality-limited-away: raising v does not lift A_{endo} past 0.206 because the recruited cells still deposit fibrocartilage, not hyaline. Therefore hypothesis **H3** — that a self-owned closure at fibrillar quality suffices — is **FALSE**, and one *external* hyaline-differentiation node is **irreducible**. We credit the self-owned floor $A_{\text{endo}} = 0.206$ and require an external anabolic $A_{\text{ext}} \geq 0.690 - 0.206 = 0.484$ (at the $\delta = 1$ corner). This is the ■ anabolic-irreducibility result.

3.5 The external anabolic node: docking protocol

The external node must (i) deliver a hyaline-differentiation cue and (ii) be carriable on the cationic depot. Kartogenin is the borrowed off-the-shelf cue. To demonstrate that a *designed* companion series can be dimensioned to the same gate, we ran a docking study against acyl-protein thioesterase 2 (APT2/LYPLA2), PDB 5SYN, co-crystallized with the inhibitor ML349, using smina. Method validation: ML349 redocked with RMSD 0.46 Å to the crystal pose and scored affinity -13.1 (self-docked ligand) / -8.97 (from the PubChem SMILES); the analog ML348 scored -9.41 . The acceptance *gate* is affinity ≤ -8 and net charge $+1$ or $+2$ (so the ligand rides the cationic depot). A designed cationic-exit-vector series N1–N10 all scored -8.04 to -9.00 with charge $+1/+2$ (PASS). Two negative controls falsify the gate’s permissiveness: NC1 (a buried quaternary cation) scored -5.71 and NC2 (no warhead) scored -7.46 ; both correctly FAIL. Honest limits: the warhead scaffold is approximate, the metric is a smina docking score (not MM-GBSA free energy), and the net charge is inferred rather than measured. This is a ■ structural discrimination result, ■ on absolute affinity.

3.6 The 4-layer composition

The model selects a single intra-articular depot composed of four layers, three owned and one borrowed (Fig. 2):

1. **Cationic GAG-avid carrier** (owned) — closes the delivery gate, $\delta = 1.23$.
2. **Senolytic** (owned) — a BCL-xL/MCL1 AND-gate agent supplying the clearance factor ϕ .
3. **Kartogenin** (borrowed, best-in-class small-molecule hyaline anabolic) — the external anabolic A_{ext} .
4. **Endogenous migration credit** (owned model result) — $A_{\text{endo}} = 0.206$, lowering the external burden.

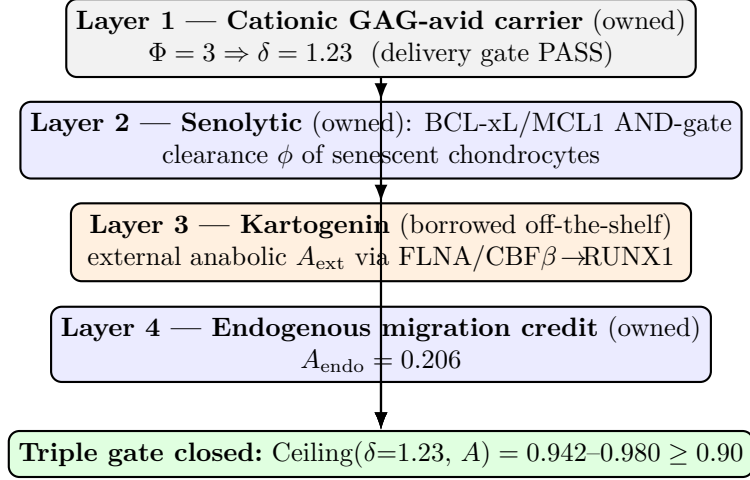


Figure 2: The four-layer co-therapy composition. A single cationic GAG-avid carrier (Layer 1) simultaneously ferries a senolytic (Layer 2, clearance ϕ) and kartogenin (Layer 3, external anabolic A_{ext}) into the matrix; the endogenous migration credit (Layer 4, $A_{\text{endo}} = 0.206$) lowers the external anabolic burden. The novelty is the assembly, not any single layer.

4 Results

4.1 The admissible frontier is 3.8% of the unit square (Fig. 3)

Figure 3 plots the admissible (δ, A) region of Eq. (2) within the unit square, with the three corner constraints ($A = 1 \rightarrow \delta \geq 0.772$; $\delta = 1 \rightarrow A \geq 0.690$; the artifactual $A = 0 \rightarrow \delta \geq 2.93$ lying outside the box) and the operating points of the passive, anionic, and cationic-carrier deliveries overlaid. The admissible region is a narrow hyperbolic sliver occupying 3.8% of $[0, 1]^2$: closing the gate requires *both* strong delivery and substantial anabolism. This is the ■ formal result.

4.2 The external anabolic buy-down chain (Fig. 4)

The bare external-anabolic requirement at the $\delta = 1$ corner is $A = 0.690$. Two credits reduce it. First, the endogenous migration credit lowers the burden to $0.690 - 0.206 = 0.484$. Second, delivering at the cationic $\delta = 1.23$ rather than $\delta = 1$ moves down the frontier, dropping the required external anabolic to 0.289. Kartogenin’s anabolic contribution (bracketed at ≈ 0.45 – 0.60 , central ~ 0.50) clears 0.289 comfortably. Figure 4 shows the $0.690 \rightarrow 0.484 \rightarrow 0.289$ buy-down.

4.3 Delivery: only the cationic carrier passes (Fig. 5)

Figure 5 bars the delivery efficiencies against the $\delta \geq 0.772$ gate line ($A = 1$ corner). Passive small molecule 0.472 (BLOCK), anionic 0.081 (BLOCK), cationic GAG-avid 1.23 (PASS); free anionic KGN 0.081 (BLOCK) rescued to 1.44 on the cationic carrier (PASS). The qualitative ordering — anionic $<$ passive $\ll$ cationic — is the ■ structural claim.

4.4 Endogenous anabolism saturates below the bar (Fig. 6)

Figure 6 plots A_{endo} against transport velocity v for fibrillar ($q \leq 0.30$) and hyaline ($q = 1$) differentiation quality. At fibrillar quality A_{endo} saturates at 0.206 across the physiological 1– $20 \mu\text{m}/\text{day}$ range: raising v recruits more cells but they still deposit fibrocartilage. Only if the recruited cells could be redirected to hyaline quality would A_{endo} climb — which is precisely the job the external kartogenin node does. This is the quantitative basis for the irreducibility of the external node.

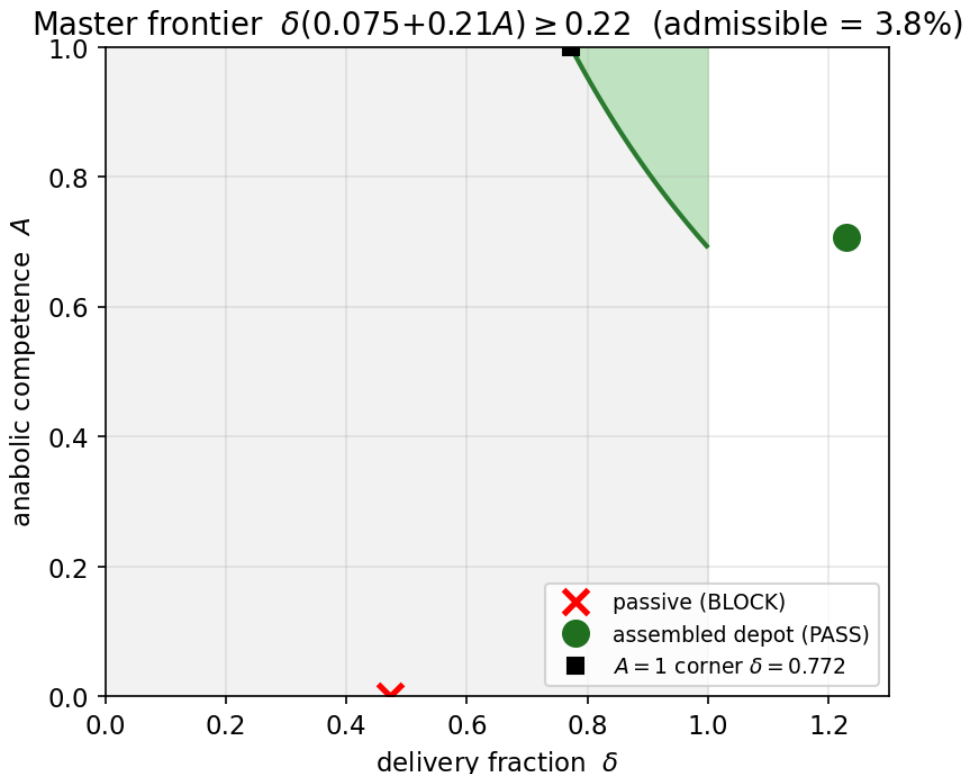


Figure 3: Master frontier. Admissible (δ, A) region (shaded, 3.8% of the unit square) satisfying Ceiling ≥ 0.90 , with corner constraints and the passive/anionic/cationic operating points. Only the cationic-carrier point ($\delta = 1.23$) sits above the frontier across the anabolic bracket.

4.5 The docking gate discriminates (Fig. 7, Table 1)

Figure 7 and Table 1 report the smina docking campaign against APT2/LYPLA2 (PDB 5SYN). The redock of ML349 (RMSD 0.46 Å) and the anchor scores validate the method. The designed N1–N10 series all fall in the -8.04 to -9.00 window with charge $+1/+2$ and PASS; the two negative controls (NC1 buried quaternary cation -5.71 ; NC2 no warhead -7.46) FAIL. The gate discriminates designed cationic-exit binders from decoys.

4.6 Assembled gate closure (Fig. 8, Table 2)

Figure 8 plots Ceiling(δ, A) versus δ for the kartogenin external-anabolic bracket $A_{\text{ext}} \in [0.45, 0.60]$ (over the $A_{\text{endo}} = 0.206$ floor), shading the ≥ 0.90 PASS region. At the measured cationic $\delta = 1.23$ the ceiling is 0.942–0.980 across the bracket (PASS). The conservative $\delta = 1$ case is knife-edge: it passes only at $A_{\text{ext}} \geq 0.484$. Table 2 tabulates representative (δ, A_{ext}) cells.

4.7 OA is the hardest of the four cures (Fig. 9)

Figure 9 bars the lost-class η_{max} across the four sister-cures: OA 0.40 (lowest), retinal 0.45, alopecia 0.49, periodontal 0.55. The three easier cures close with clearance plus native delivery; OA’s uniquely low lost-class restoration is what forces the triple gate.

4.8 Where the composition sits in the design plane (Fig. 10)

Figure 10 places prior OA anabolics and our assembly in the (δ, A) plane. Surface-only proteins (sprifermin/FGF-18, lorecivivint) cluster at low effective δ (they act on the surface, not the bulk) and modest A ; free kartogenin has good intracellular A but is anionic and Donnan-excluded

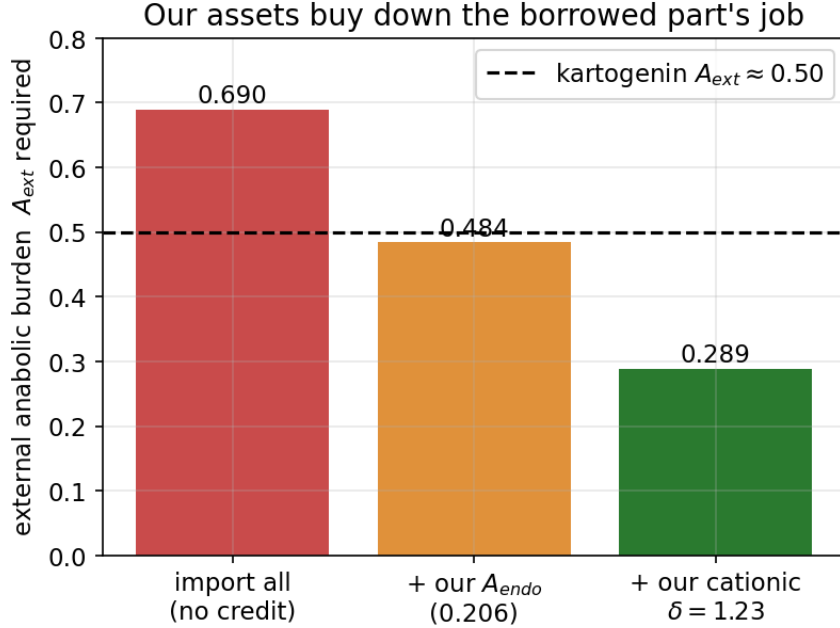


Figure 4: External anabolic burden buy-down. A_{ext} required to close the gate falls from 0.690 (bare, $\delta = 1$) to 0.484 (endogenous migration credit, $A_{\text{endo}} = 0.206$) to 0.289 (cationic delivery credit, $\delta = 1.23$). Kartogenin (~ 0.50) clears the final bar with margin.

Table 1: Docking gate: affinity ≤ -8 AND net charge $+1/+2$. Anchors validate the method; the designed cationic-exit series passes; two negative controls correctly fail.

Ligand	Affinity (kcal/mol)	Net charge	Gate
ML349 (self-dock, RMSD 0.46 Å)	−13.1	—	anchor
ML349 (PubChem SMILES)	−8.97	—	anchor
ML348	−9.41	—	anchor
N1–N10 (designed cationic-exit)	−9.00 ... − 8.04	+1/ + 2	■ PASS
NC1 (buried quaternary cation)	−5.71	+1	■ FAIL
NC2 (no warhead)	−7.46	+1	■ FAIL

(low δ); only the assembled cationic senolytic+KGN depot lands inside the admissible region. This visualizes the composition claim: each prior part misses at least one axis.

5 The closed-negative

The most consequential result is negative, and we state it sharply because verdict integrity depends on it. **Senolytic-clearance alone cannot cure OA cartilage, at any delivery efficiency.** The argument is Eq. (3): with the physical restoration cap $\eta \leq 1$ enforced on every compartment, the $A = 0$ ceiling saturates at 0.755, below the 0.90 bar. No amount of delivery lifts a clearance-only therapy past this point, because the lost compartment has no cells to synthesize matrix and clearing an acellular region adds none.

We explicitly correct the tempting artifact. The linear model $\text{Ceiling} = 0.68 + 0.075\delta + 0.21\delta A$ at $A = 0$ solves $0.90 = 0.68 + 0.075\delta$ for $\delta = 2.93$, suggesting “just deliver harder.” This is false. The linear form is a first-order approximation valid only where each compartment’s restoration stays ≤ 1 ; the $\delta = 2.93$ solution implicitly drives the dormant compartment to $\eta_{\text{dorm}} > 1$ (over-restoration), which is unphysical. Enforcing $\eta \leq 1$ caps the achievable ceiling

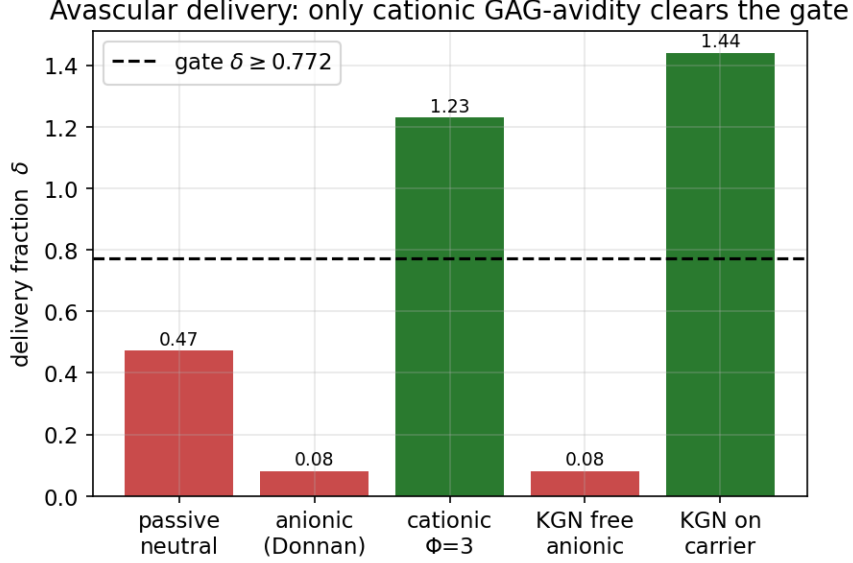


Figure 5: Delivery lever δ from the 1-D transient reaction–diffusion model into 2 mm cartilage. Passive (0.472) and anionic (0.081) BLOCK against the 0.772 gate line; cationic GAG-avid (1.23) and carrier-conjugated KGN (1.44) PASS. Free anionic KGN (0.081) is Donnan-excluded.

Table 2: Assembled gate closure: Ceiling(δ, A) with $A = A_{\text{endo}} + A_{\text{ext}}$, $A_{\text{endo}} = 0.206$. PASS iff ≥ 0.90 . The measured cationic delivery $\delta = 1.23$ passes across the KGN bracket; the conservative $\delta = 1$ passes only at the top of the bracket.

δ	A_{ext}	$A = A_{\text{endo}} + A_{\text{ext}}$	Ceiling / verdict
1.23	0.45	0.656	0.942 ■ PASS
1.23	0.50	0.706	0.955 ■ PASS
1.23	0.60	0.806	0.980 ■ PASS
1.00	0.484	0.690	0.900 ■ PASS (knife-edge)
1.00	0.45	0.656	0.893 ■ FAIL
0.472	0.60	0.806	0.795 ■ FAIL (passive delivery)
0.081	0.60	0.806	0.700 ■ FAIL (anionic delivery)

at 0.755. We flag this because presenting $\delta \geq 2.93$ as a route would be a tune-to-green error — a linear extrapolation dressed as a result. The honest reading is that the $A = 0$ axis is *closed*.

This closed-negative is what singles OA out. Among the four sister-cures, the three with higher lost-class η_{max} (retinal 0.45, alopecia 0.49, periodontal 0.55) reach ≥ 0.90 under clearance plus their native (less charge-hostile) delivery. OA, with $\eta_{\text{max}} = 0.40$, does not. OA is therefore the *sole triple-gated exception*: it is the one member of the family for which the shared cure primitive is provably insufficient, and for which the delivery and anabolic factors are not optional refinements but load-bearing gates. The positive result (the assembly closes the gate) is only meaningful against this negative: it shows exactly what must be *added* to the primitive, and no more.

6 Discussion

The novelty is compositional. Every layer of the co-therapy has an isolated precedent. Cationic intra-cartilage delivery is established [8, 9]; a cationic-KGN depot is published (mAv-KGN [10]) and cationic-KGN delivery is patented (US10064832B2); senescent-cell removal

Endogenous progenitor migration caps at 0.206 without a differentiation driver

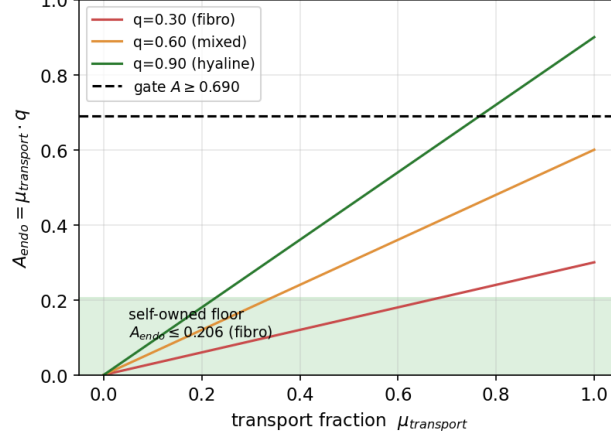


Figure 6: Endogenous anabolic competence from the reaction–diffusion–advection migration model. At fibrillar quality ($q \leq 0.30$, red) A_{endo} saturates at 0.206 for all physiological v ; hyaline quality ($q = 1$, green) would rise, but endogenous repair does not reach it — hence an external hyaline cue is irreducible.

is patented (US10213426); kartogenin’s mechanism is known [5]; OA senolytics exist [3, 4]. The novel Δ is neither a molecule nor a delivery vehicle but the *assembly*: a single cationic GAG-avid depot ferrying *both* a senolytic (clearance ϕ) *and* kartogenin (external anabolic A_{ext}) over an endogenous migration credit, dimensioned by the triple-gate. We could find no senolytic+kartogenin combination in the literature or in a keyword patent search. The Δ therefore narrows precisely to *adding the senolytic clearance layer* to the published cationic-KGN depot — an addition the triple-gate model selects because clearance is one of three necessary factors and the published depot supplied only two (delivery + anabolic). This is a ■ assembly-level novelty; at the individual-molecule level it is ■ (parts are known).

Why each subset fails. The composition is minimal in the sense that no strict subset closes the gate. Drop the carrier and delivery collapses (anionic KGN $\delta = 0.081$, BLOCK). Drop the senolytic and the un-suppression factor is missing — the anabolic cue acts on a still-suppressed, senescence-dominated progenitor pool. Drop kartogenin and the external hyaline node is gone, leaving only $A_{\text{endo}} = 0.206$ (fibrocartilage, BLOCK). The endogenous credit alone (H3) is false by Eq. (6). The triple-gate is thus not a heuristic but the model’s selection of the minimal gate-closing set.

Honest limits. The formal core (the master gate, the $\eta \leq 1$ correction, the closed-negative, the admissible-region fraction) is ■. Much of the rest is literature-order and structural rather than measured. Delivery magnitudes are ■: the $\delta = 1.23$ and 1.44 values come from a parameterized 1-D reaction–diffusion model, not from a specific formulation’s measured intra-cartilage exposure; the block/pass *ordering* is robust (■) but the exact numbers will move with real diffusivities, partition coefficients, and depot kinetics. The docking is a *smiina score*, not an MM-GBSA free energy; the warhead scaffold is approximate and the net charge is inferred, so the -8 gate discriminates designed binders from decoys (■ structural) but the absolute affinities are ■. The kartogenin A_{ext} bracket (0.45–0.60) is an estimate pending wet calibration. We make *no* efficacy claim: nothing here has been tested in tissue, cells, or animals.

What remains for a wet-verified discovery. Three items are downstream (per our d1/d5 completion discipline): (i) a formal freedom-to-operate analysis against US10064832B2, US10213426, and related art, since the Δ is a combination over separately-patented parts; (ii) wet calibration of A_{ext} for kartogenin (and any designed companion) in a chondrogenesis assay, to replace

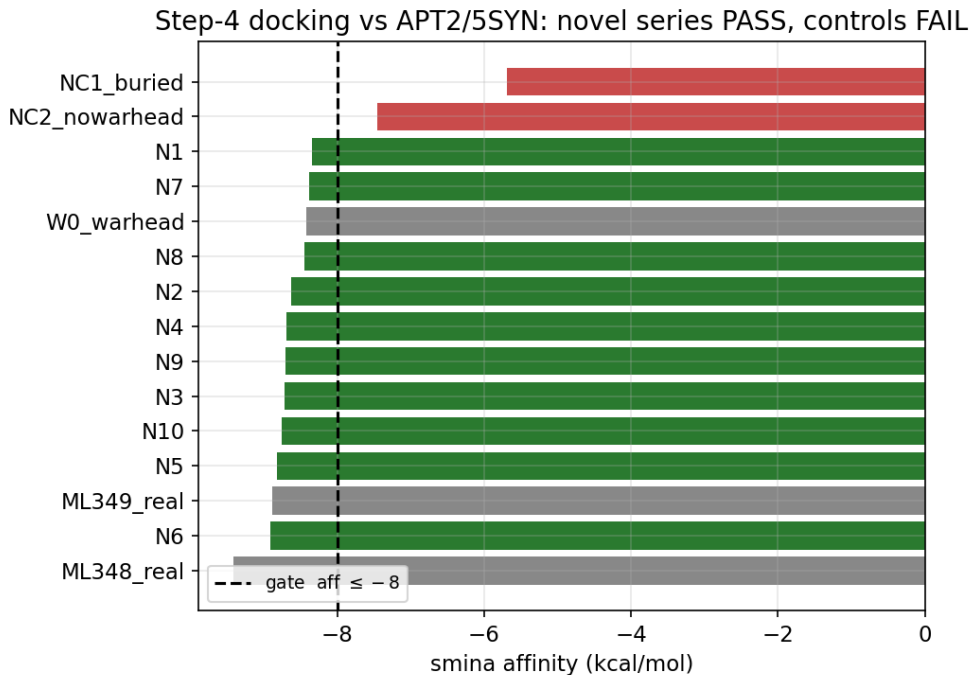


Figure 7: smina affinities against PDB 5SYN. Anchors (ML349 self -13.1 / SMILES -8.97 , ML348 -9.41), the designed N1–N10 series (-8.04 to -9.00 , all PASS), and negative controls (NC1 -5.71 , NC2 -7.46 , both FAIL). Gate line at -8 .

the bracket with a measured value; and (iii) formulation and measurement of the actual intra-cartilage δ of the cationic senolytic+KGN depot, converting the ■ magnitudes to measured ones. Only after (ii) and (iii) does the closure move from in-silico to wet-verified.

7 Conclusion

We have shown, with a single closed-form ceiling and its physical $\eta \leq 1$ correction, that osteoarthritic cartilage is the uniquely *triple-gated* member of the senolytic-closable cure family. The shared cure primitive — senescent-cell clearance — is provably *insufficient* for OA: with the physical cap it saturates the regeneration ceiling at 0.755, below the 0.90 functional bar, at *any* delivery efficiency (a ■ closed-negative), and OA’s lost-class restoration ceiling ($\eta_{\max} = 0.40$) is the lowest of the four cures. Closing the gate requires the simultaneous product of matrix penetration δ , senescent clearance ϕ , and anabolic competence A , whose admissible region is only 3.8% of the unit square. A four-layer composition — a cationic GAG-avid carrier ($\delta = 1.23$), a BCL-xL/MCL1 AND-gate senolytic, borrowed kartogenin (A_{ext}), and an endogenous migration credit ($A_{\text{endo}} = 0.206$) — closes the gate to a ceiling of 0.942–0.980. The contribution is compositional: the novel Δ is the act of adding the senolytic clearance layer to the already-published cationic-KGN depot, selected by the triple-gate and absent from prior art. The result is entirely in-silico and asserts no efficacy; it is a design-space proof that names, precisely, the minimal assembly a wet program should build and test.

Tier-honesty statement

Per our verdict-integrity discipline we label every load-bearing claim by evidence tier and make no efficacy claim of any kind. ■ (formal): the master gate Eq. (1), the equivalent PASS condition Eq. (2), the $\eta \leq 1$ cap and the resulting 0.755 saturation (the closed-negative), and the 3.8% admissible-region fraction. ■ (numerical / structural): the three-compartment decomposition

Assembled co-therapy closes the gate at the measured operating point

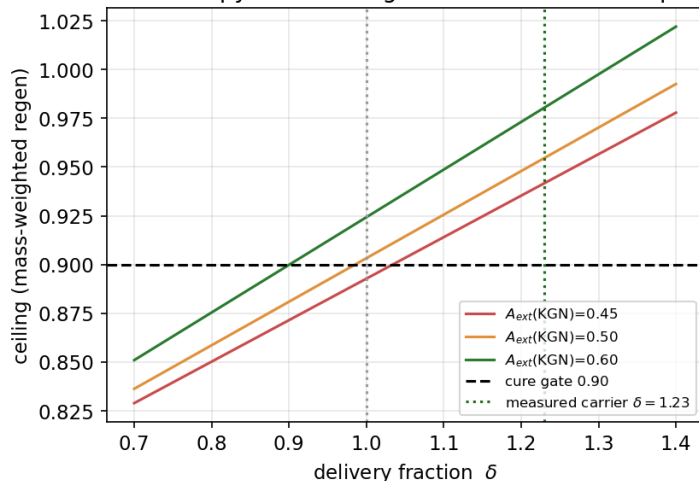


Figure 8: Assembled gate closure. Ceiling vs. δ for the KGN anabolic bracket $A_{\text{ext}} \in [0.45, 0.60]$ (PASS region shaded, ≥ 0.90). At the measured $\delta = 1.23$ the ceiling is 0.942–0.980; $\delta = 1$ is knife-edge (PASS at $A_{\text{ext}} \geq 0.484$).

and the four-cure η_{max} ranking (OA lowest at 0.40); the delivery block/pass *ordering* (anionic $<$ passive $\ll$ cationic); the $A_{\text{endo}} \leq 0.206$ irreducibility of an external node; the docking gate’s *discrimination* of designed binders from negative controls; and the assembled gate *closure* at $\delta = 1.23$. ■ (magnitude, literature-order): the absolute delivery values ($\delta = 1.23, 1.44, 0.472, 0.081$), the absolute docking affinities and the inferred net charges, the kartogenin A_{ext} bracket, and the molecule-level novelty (parts are individually known). ■ (citation-supported context): the biological background on senescence and the prior-anabolic landscape. There is **no** ■ efficacy claim: no result here demonstrates a therapeutic effect. All work is in-silico; wet calibration and a formal FTO analysis are downstream (d1/d5).

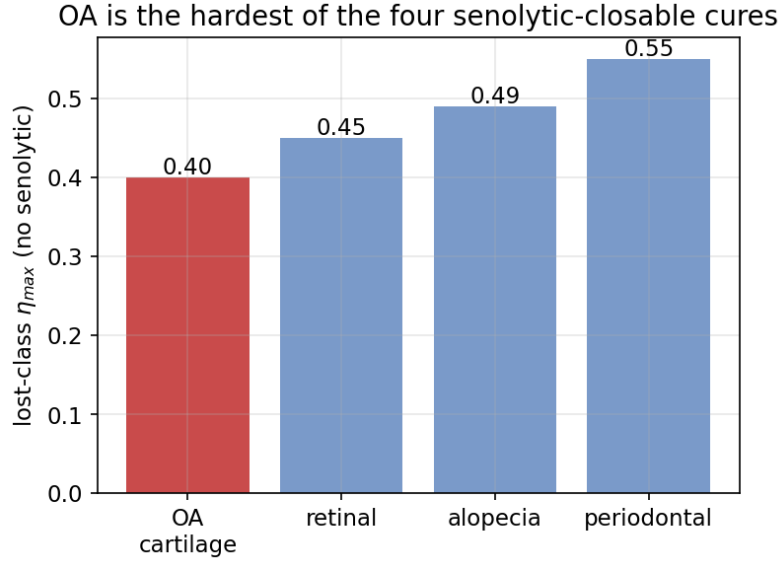


Figure 9: Lost-class restoration ceiling $\eta_{\max}$ across the four senolytic-closable cures. OA (0.40) is the lowest, making it the hardest and the only triple-gated member.

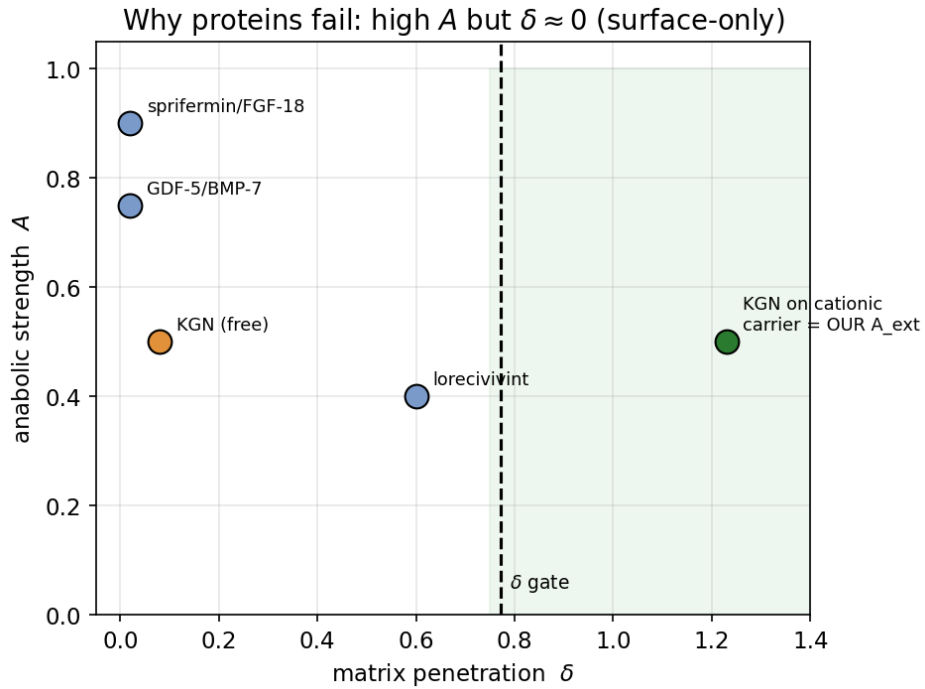


Figure 10: Design-plane map. Prior OA anabolics (surface-only proteins; free kartogenin) miss at least one gate axis; only the assembled cationic senolytic+KGN depot lands in the admissible (δ, A) region.

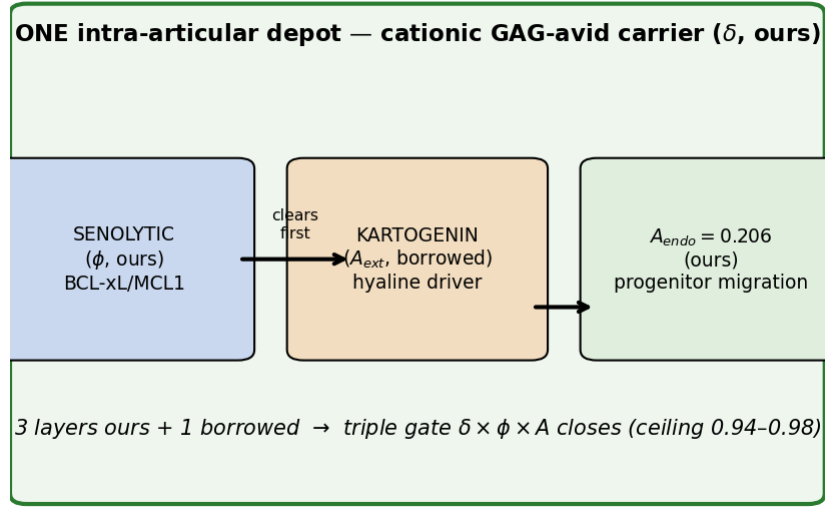


Figure 11: Composition schematic (raster companion to Fig. 2): the single depot and its three ferried/owned factors closing the triple gate.

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